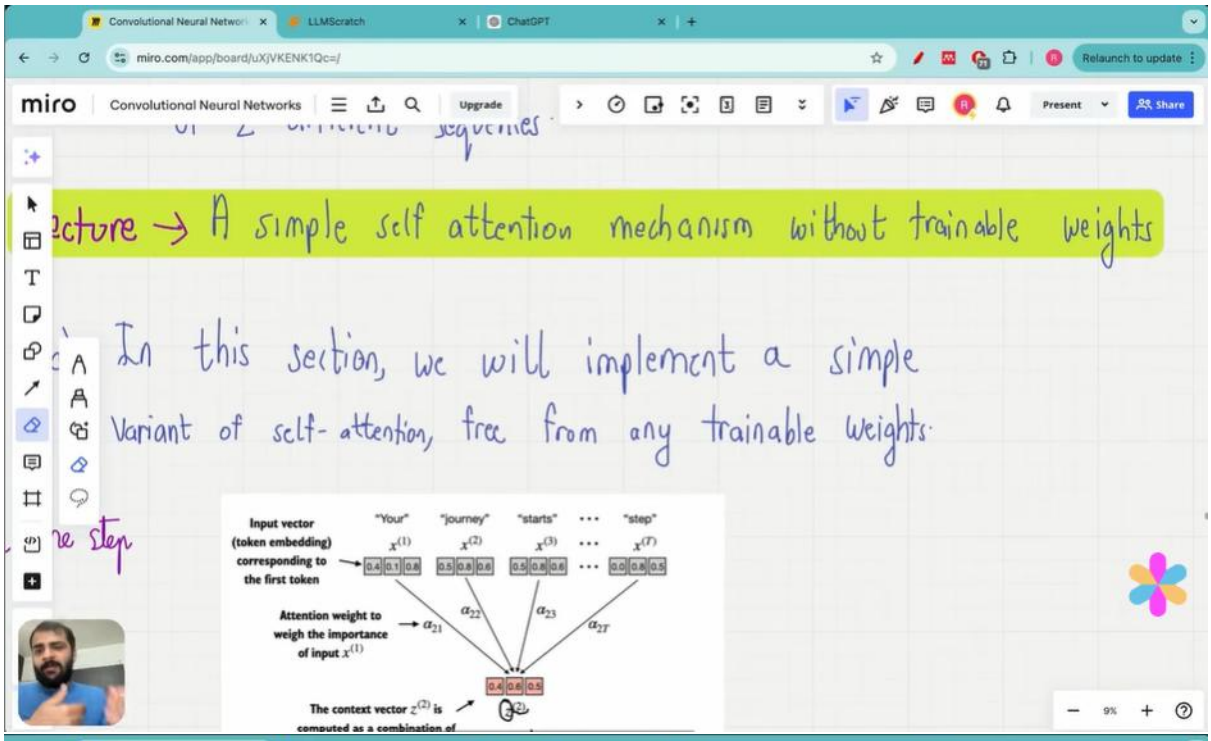
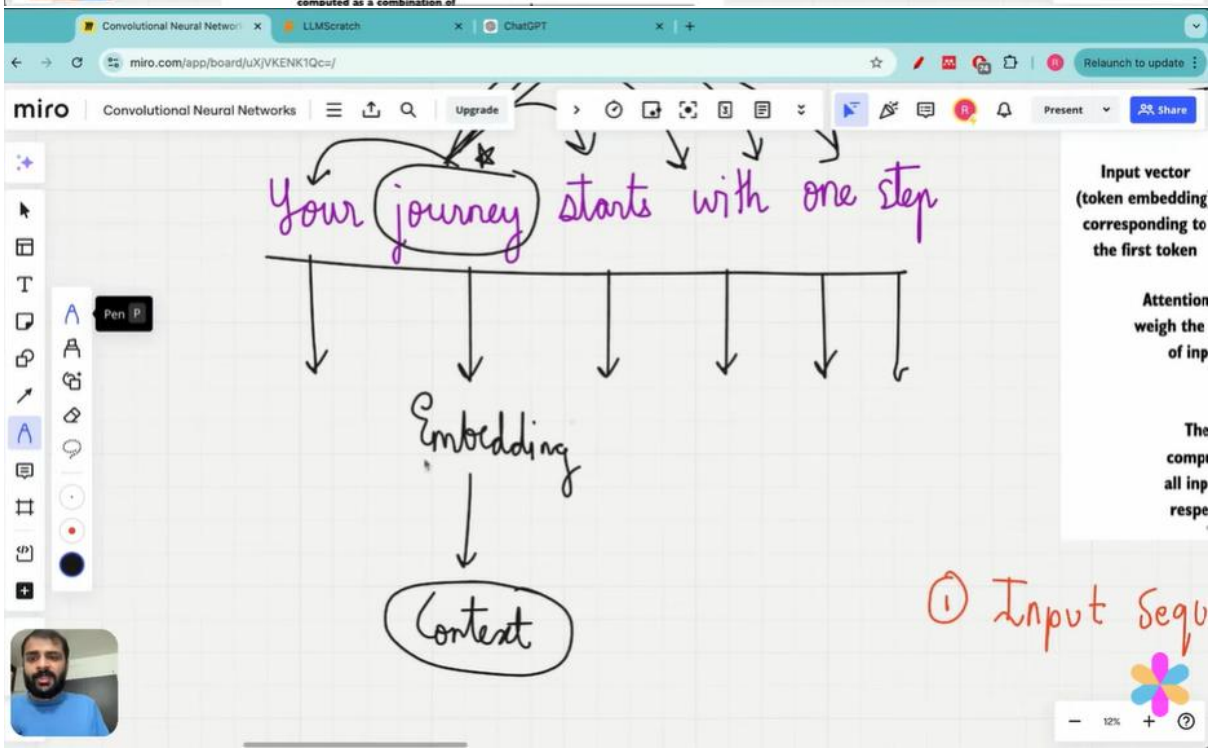
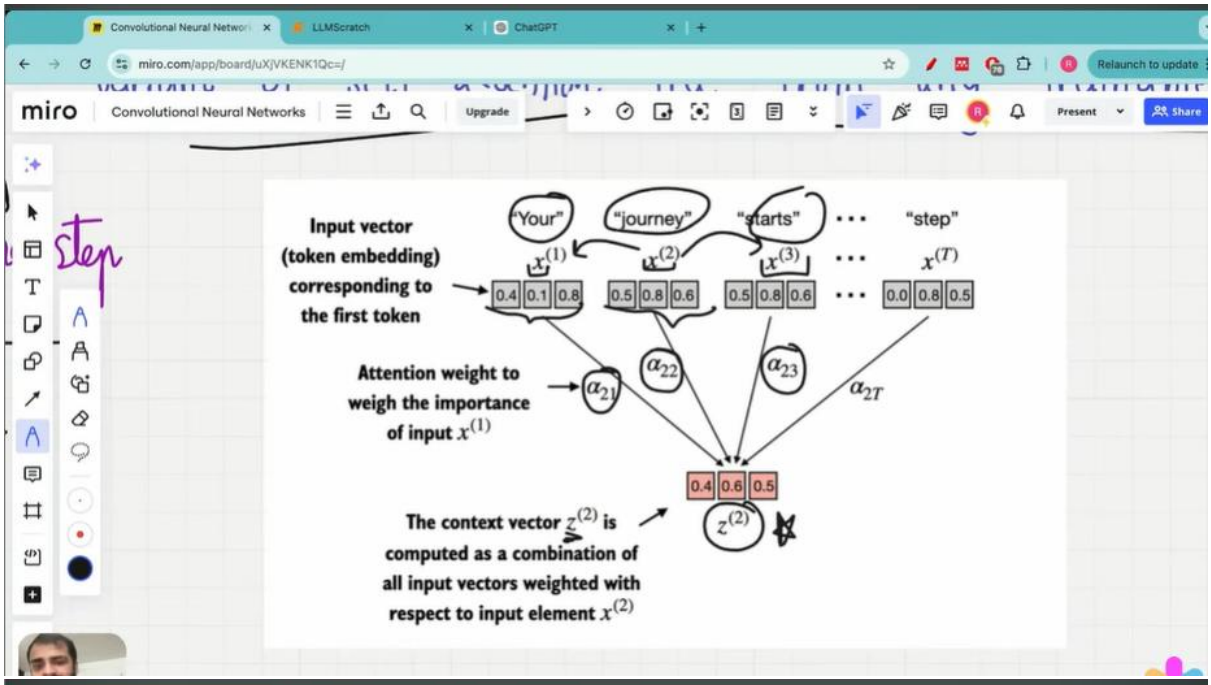
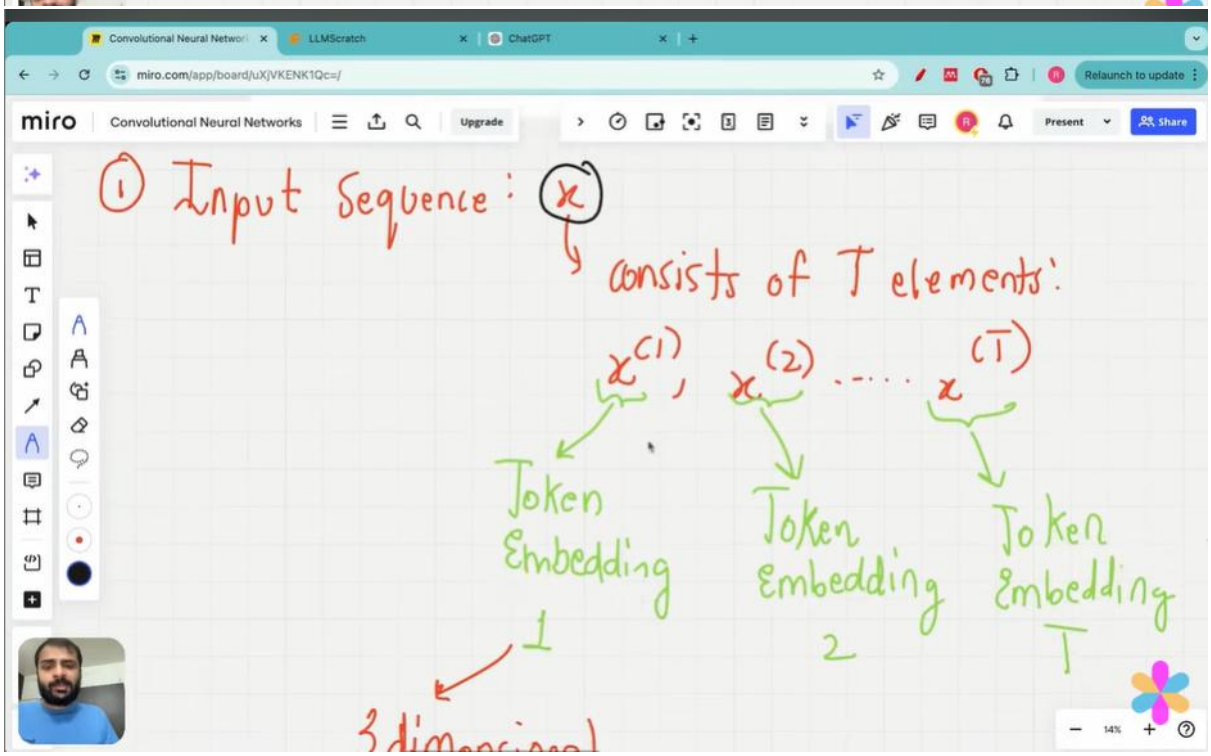
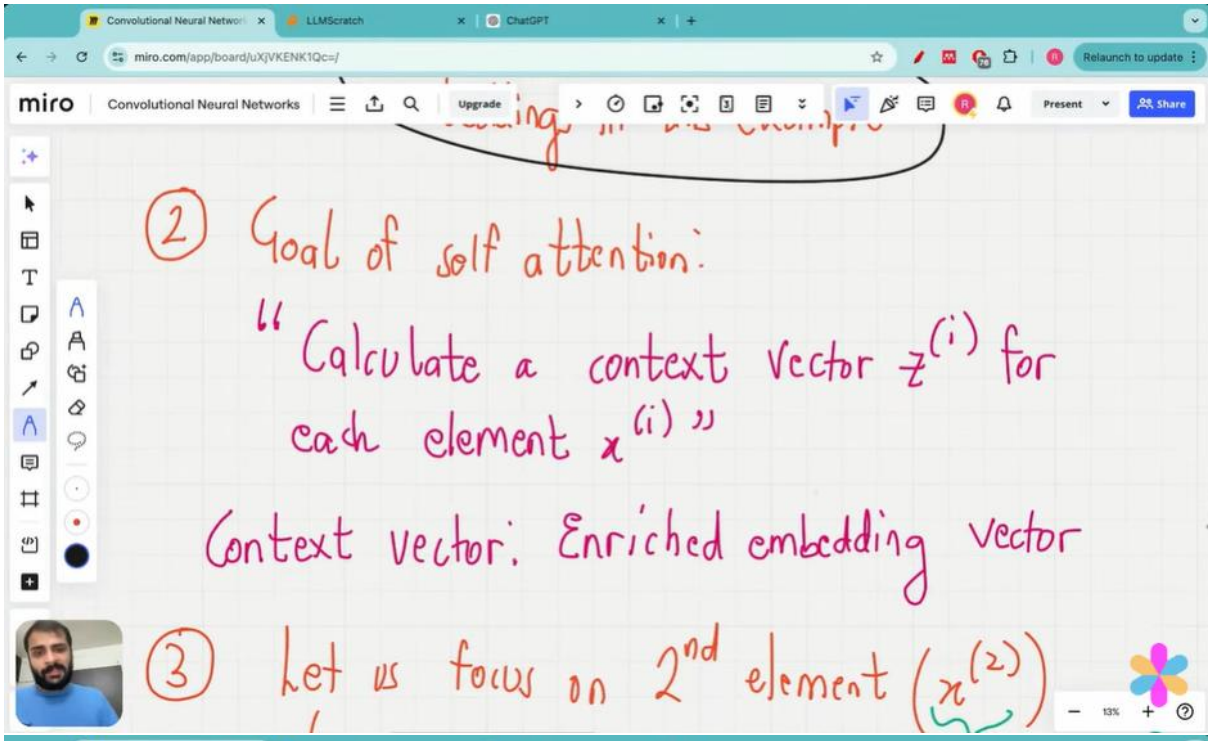
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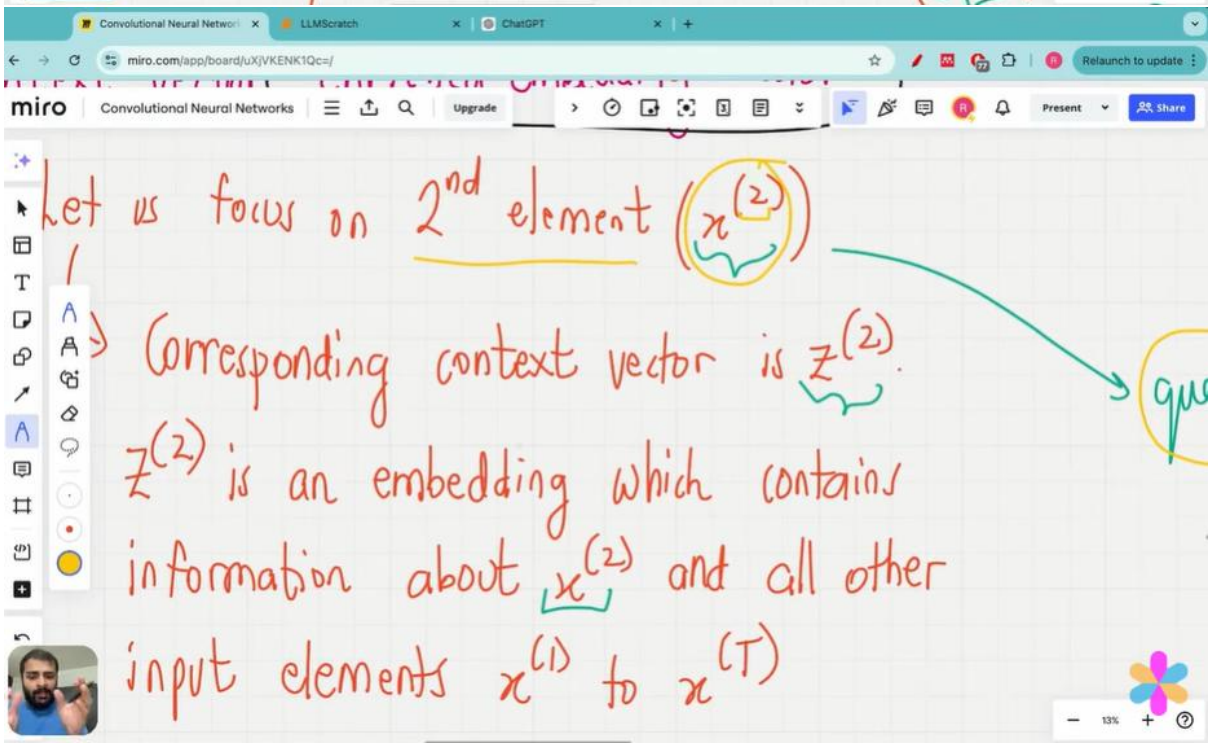

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ecture → A simple self attention mechanism without trainable weights
In this section, we will implement a simple
variant of self-attention, free from any trainable weights.
Input vector (token embedding) corresponding to the first token
"Your"
"journey"
"starts"
"step"
Attention weight to weigh the importance of input $x^{(1)}$
The context vector $z^{(2)}$ is computed as a combination of

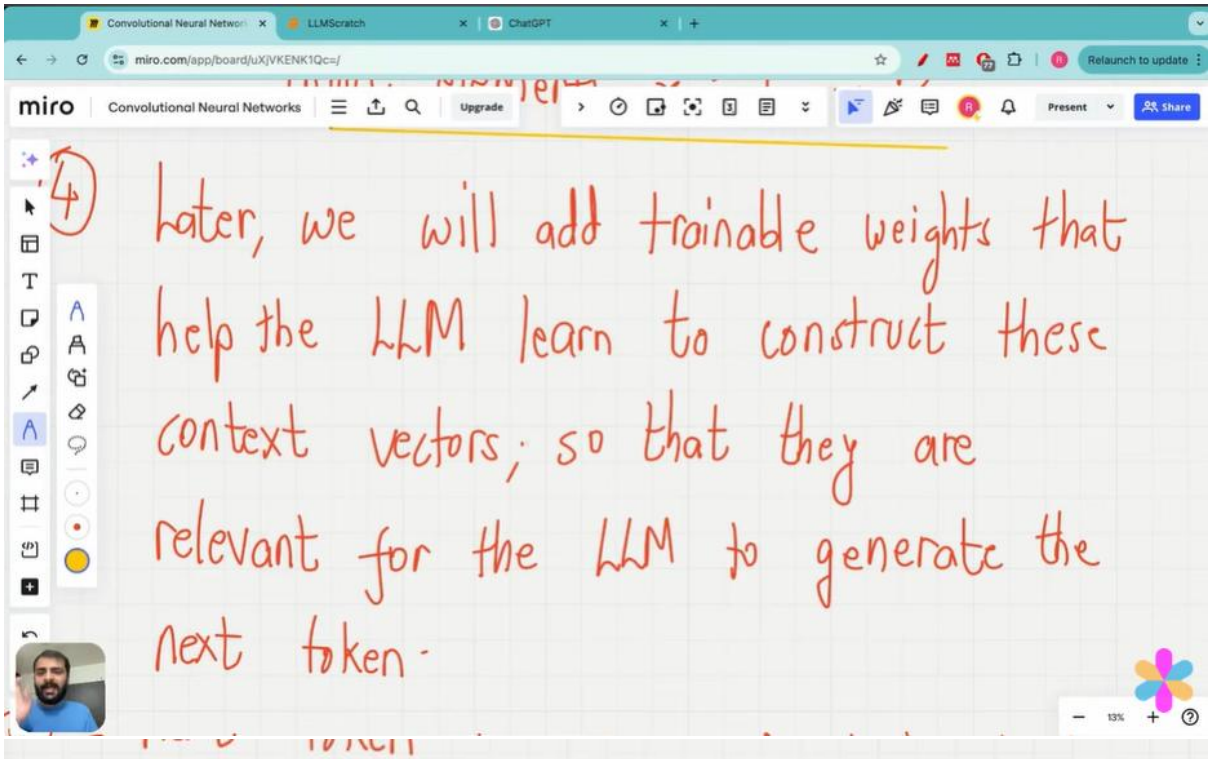

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Pen P
Your journey starts with one step
Embedding
Context
Input vector (token embedding corresponding to the first token
Attention weigh the of inp
The compu all inp respe
① Input Sequ


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Input vector (token embedding) corresponding to the first token
"Your"
"journey"
"starts"
"step"
$x^{(1)}$
$x^{(2)}$
$x^{(3)}$
$x^{(T)}$
0.4 0.1 0.8
0.5 0.8 0.6
0.5 0.8 0.6
0.0 0.8 0.5
Attention weight to weigh the importance of input $x^{(1)}$
α_{21}
α_{22}
α_{23}
α_{2T}
0.4 0.6 0.5
$z^{(2)}$
The context vector $z^{(2)}$ is computed as a combination of all input vectors weighted with respect to input element $x^{(2)}$

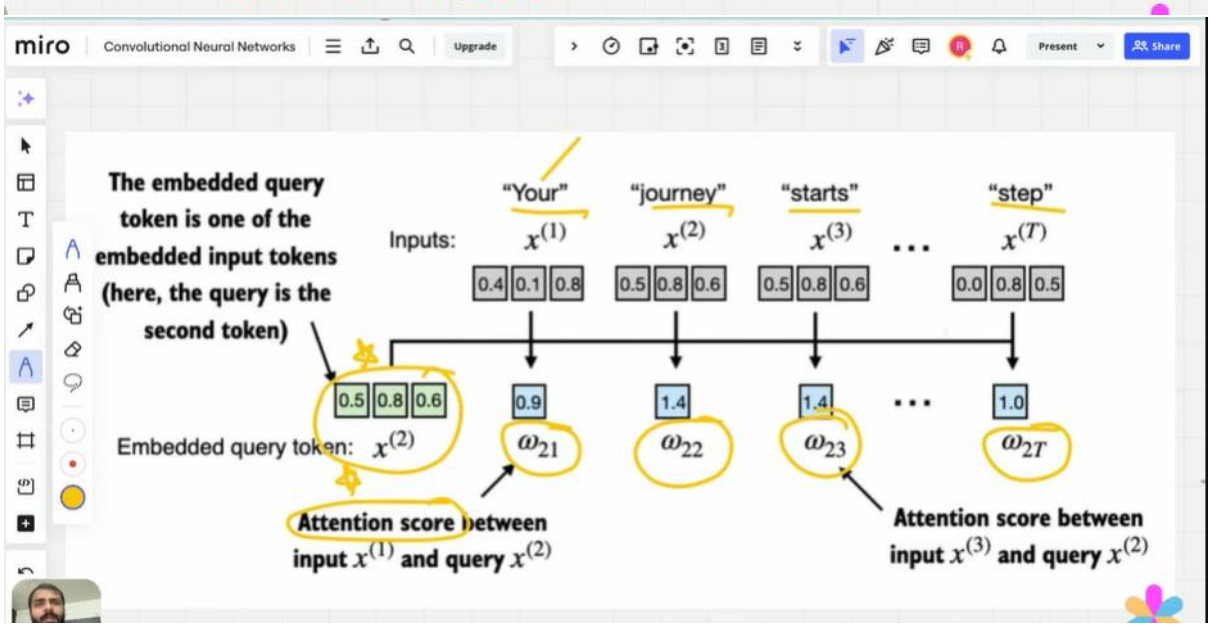

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① Input Sequence: x
consists of T elements:
$x^{(1)}$, $x^{(2)}$ ..... $x^{(T)}$
Token Embedding 1
Token Embedding 2
Token Embedding T
3 dimensional
14%

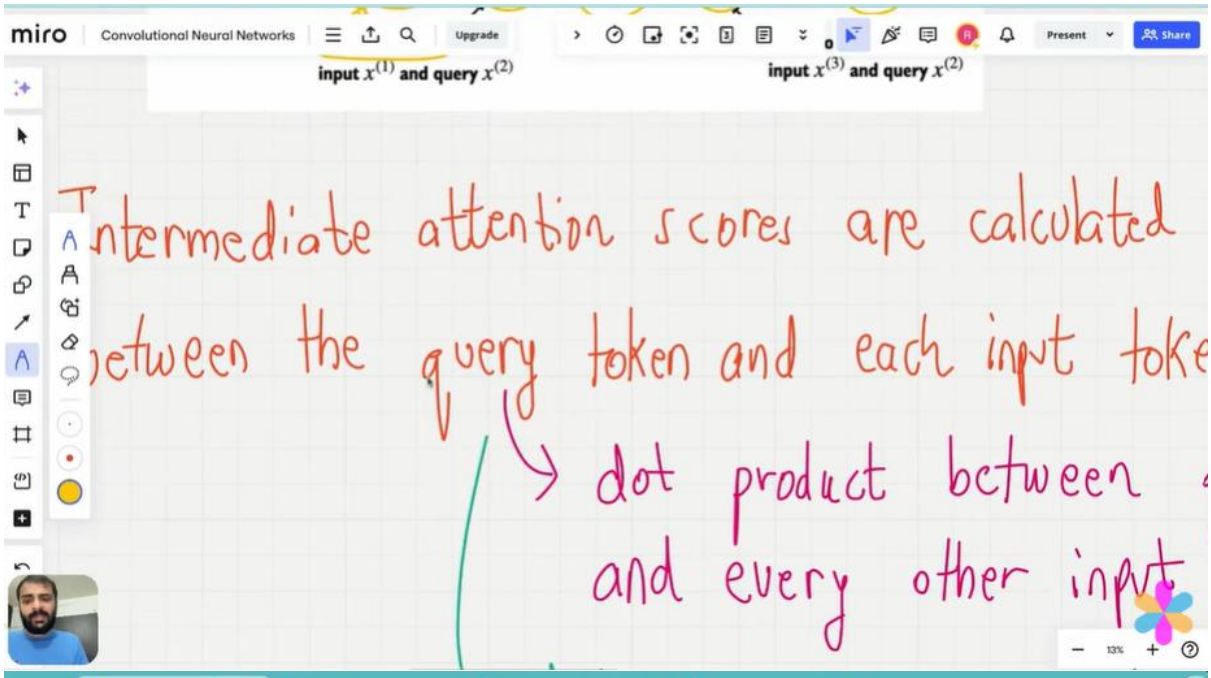

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② Goal of self attention:
" Calculate a context vector $z^{(i)}$ for each element $x^{(i)}$ "
Context vector: Enriched embedding vector
③ Let us focus on 2nd element $(x^{(2)})$
13%

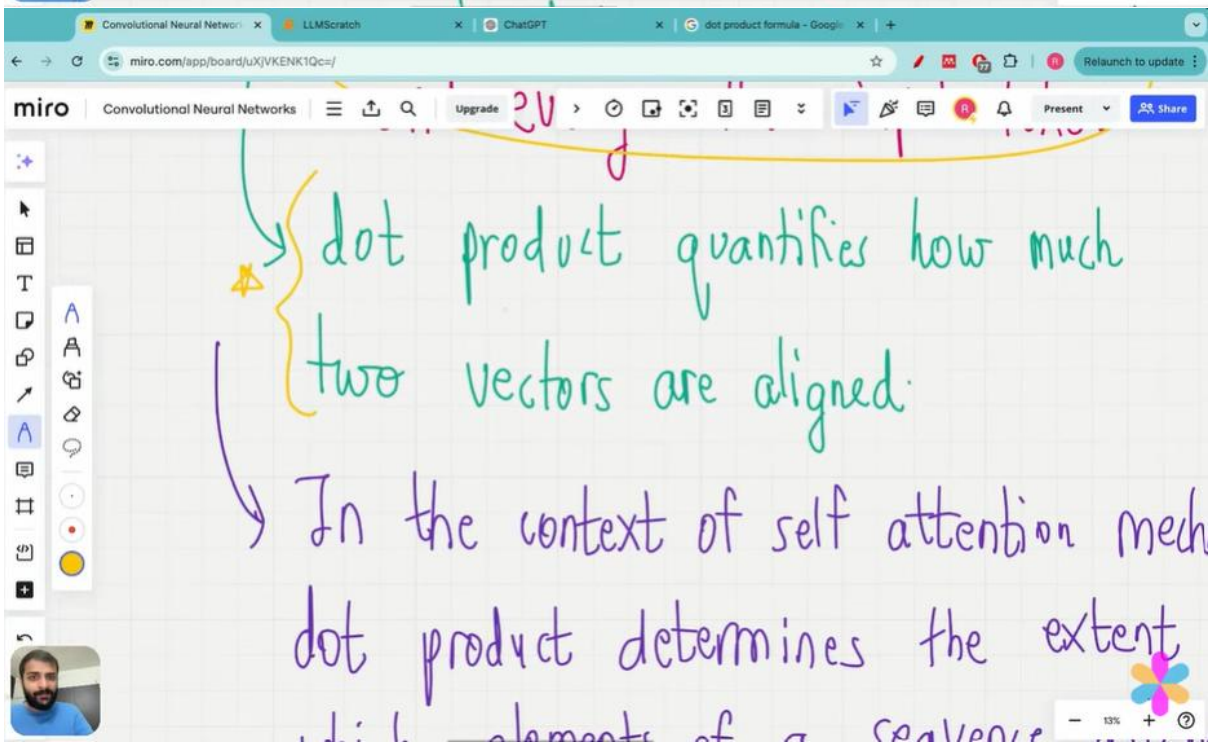

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Let us focus on 2nd element $(x^{(2)})$
Corresponding context vector is $z^{(2)}$.
$z^{(2)}$ is an embedding which contains information about $x^{(2)}$ and all other input elements $x^{(1)}$ to $x^{(T)}$
13%

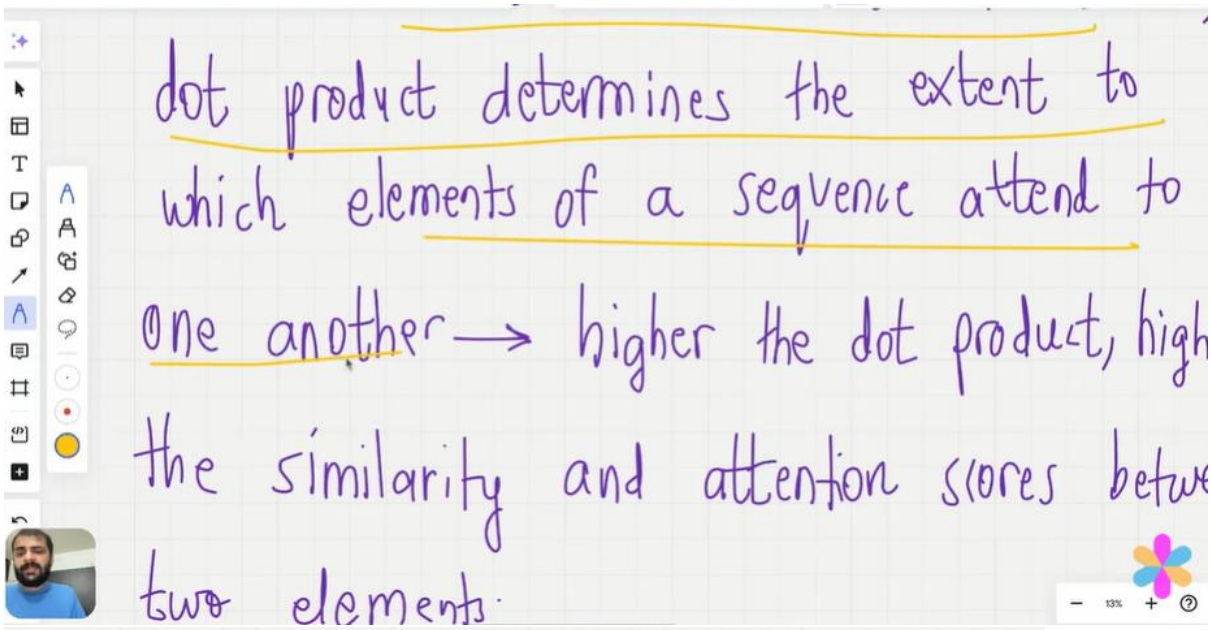


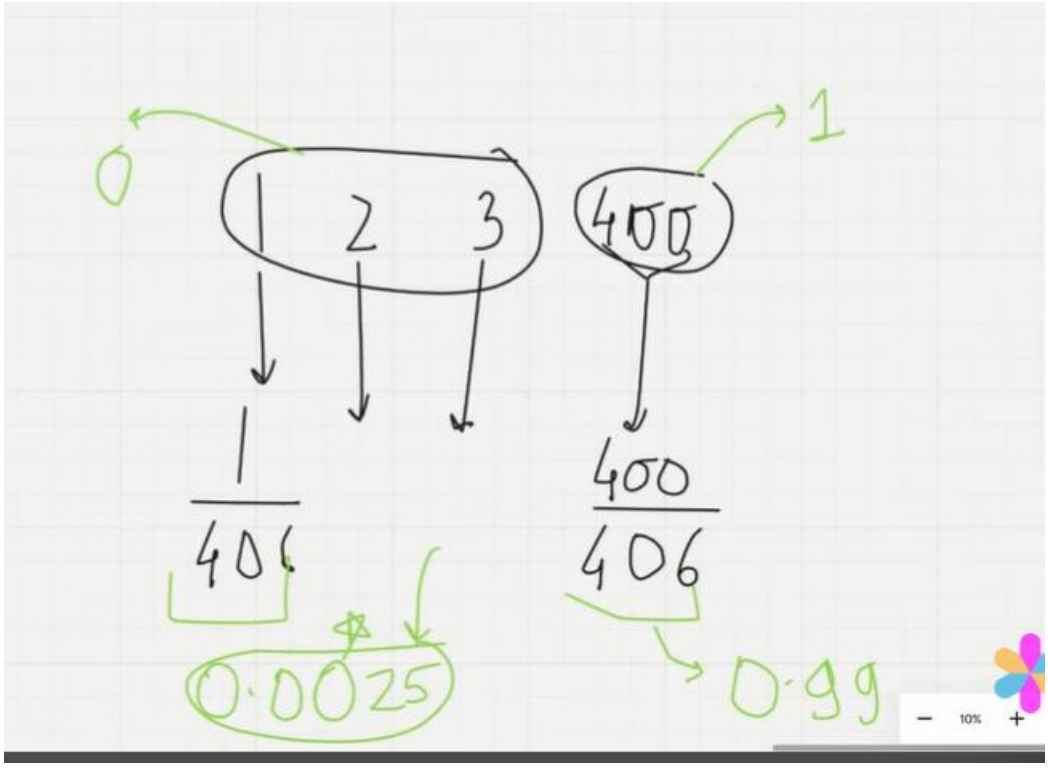
5) First step of implementing self-attention is to compute the intermediate values w, also referred to as attention scores.



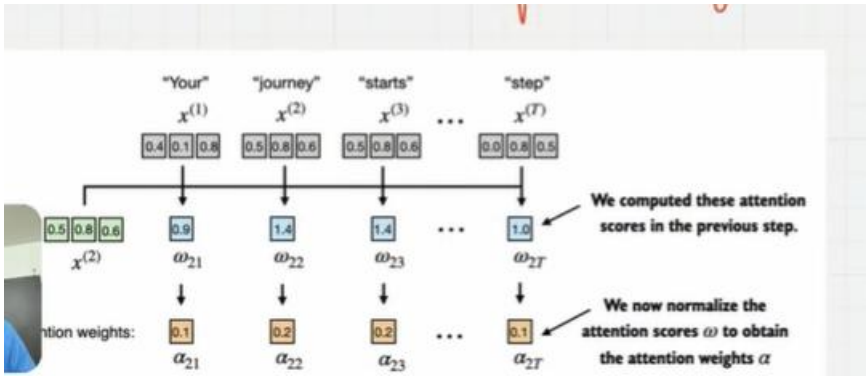

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input $x^{(1)}$ and query $x^{(2)}$
input $x^{(3)}$ and query $x^{(2)}$
Intermediate attention scores are calculated
between the query token and each input toke
dot product between
and every other input

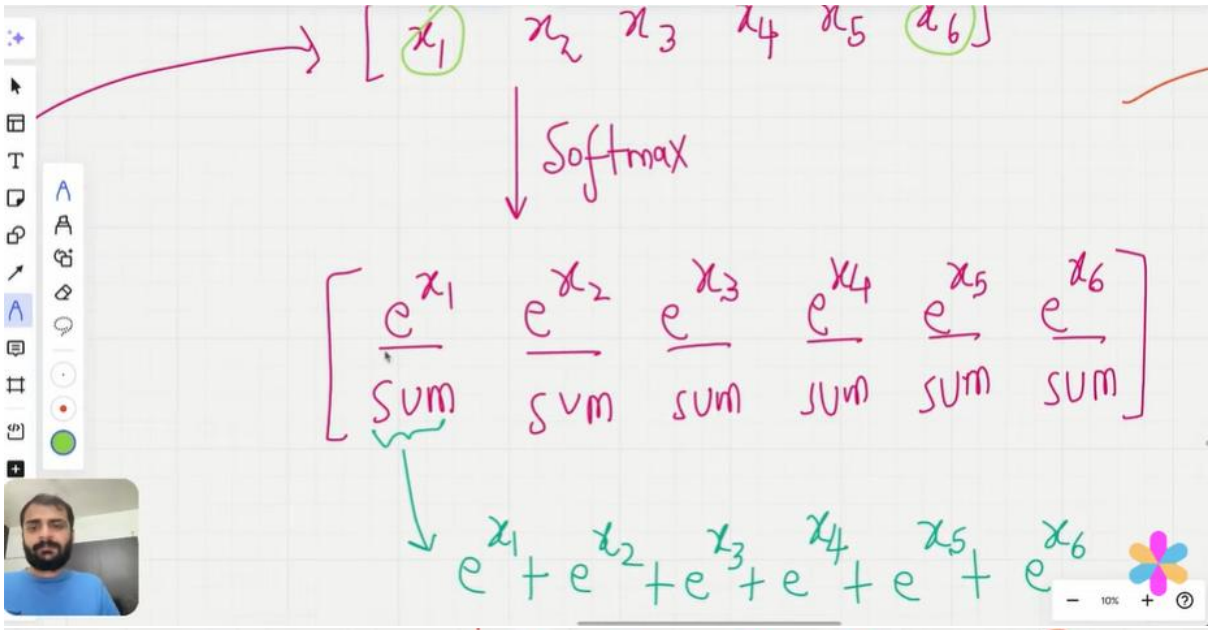

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dot product formula - Google
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dot product quantifies how much
two vectors are aligned.
In the context of self attention mech
dot product determines the extent

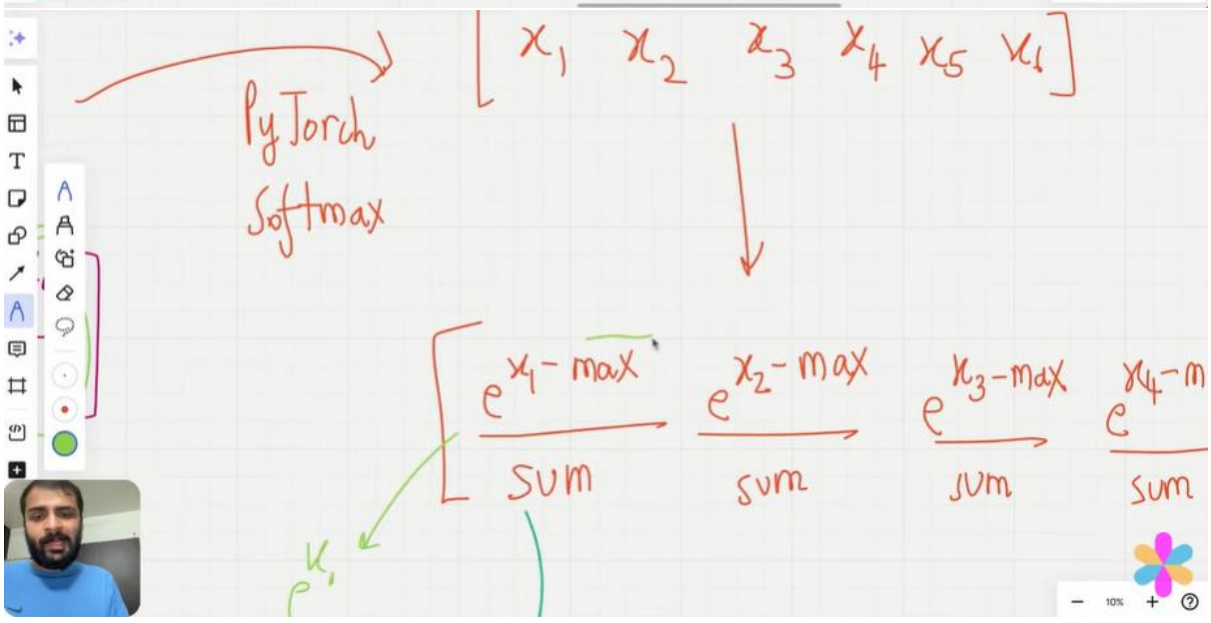




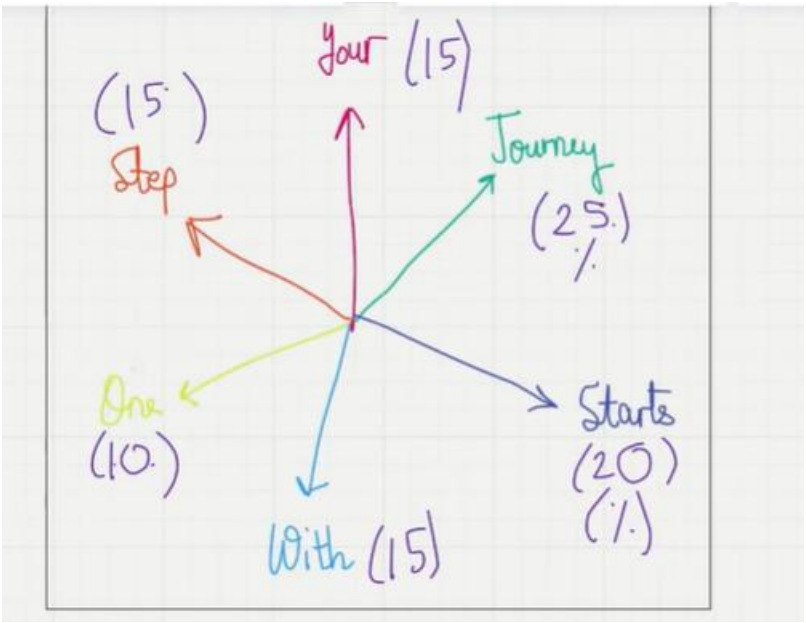
Why normalization to 1 is not suitable , we add all the attention scores divide each of them by sum to normalize it

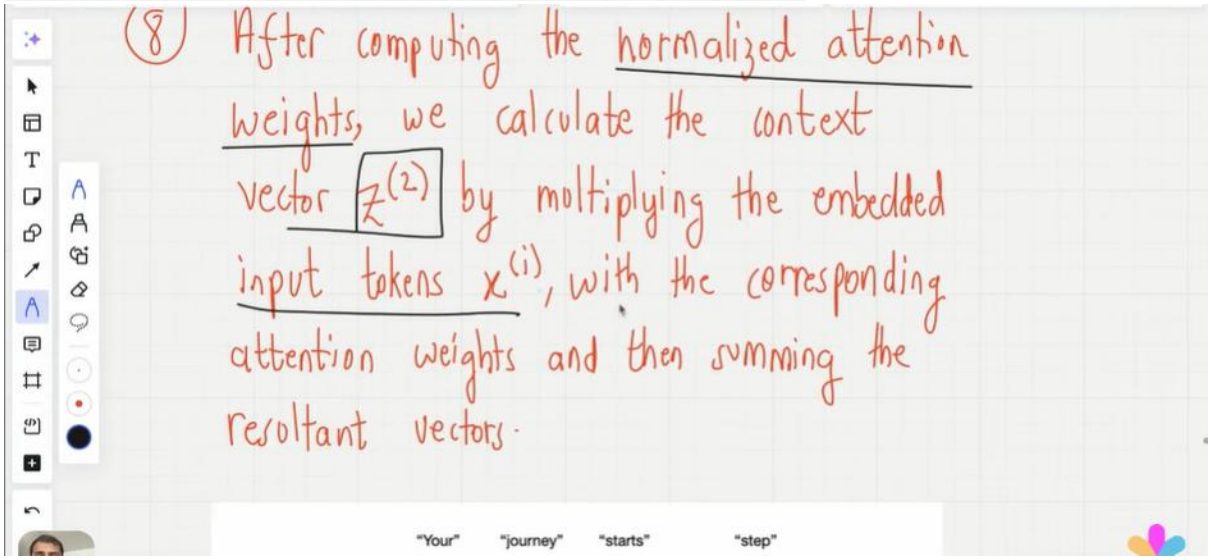




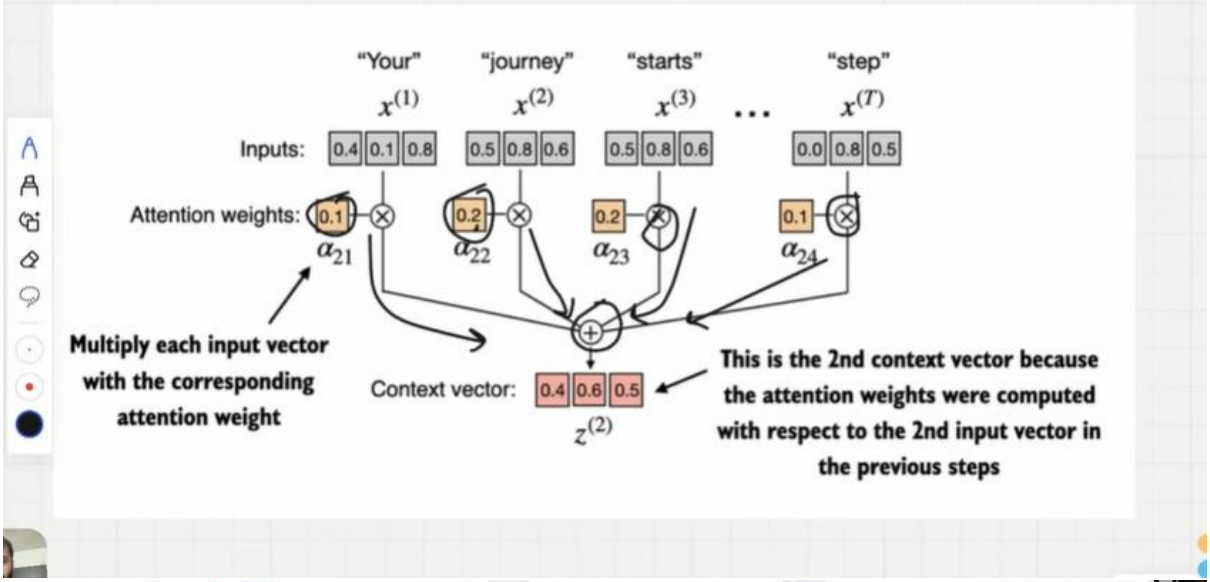


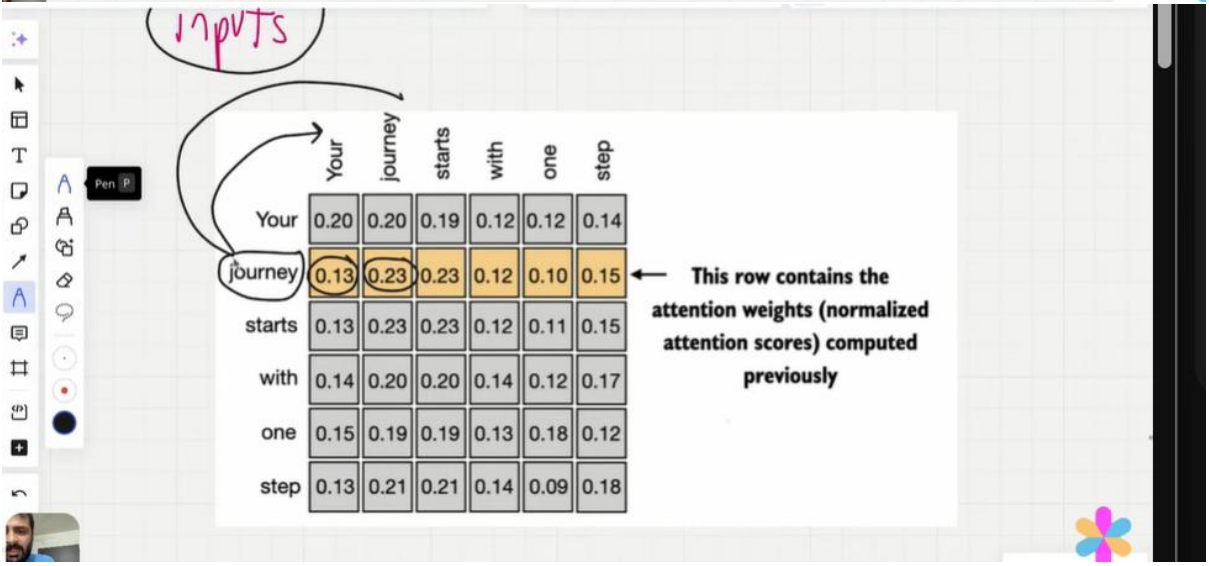
https://chatgpt.com/share/6aec0b20-5580-83ee-a483-740c62c55c3f





https://chatgpt.com/share/6aec1216-0868-83e8-9cd5-56ed2528995b


"Your"
"journey"
"starts"
"step"
$x^{(1)}$
$x^{(2)}$
$x^{(3)}$
$x^{(T)}$
Inputs:
0.4 0.1 0.8
0.5 0.8 0.6
0.5 0.8 0.6
0.0 0.8 0.5
Attention weights:
0.1
0.2
0.2
0.1
α_{21}
α_{22}
α_{23}
α_{24}
Multiply each input vector with the corresponding attention weight
Context vector:
0.4 0.6 0.5
$z^{(2)}$
This is the 2nd context vector because the attention weights were computed with respect to the 2nd input vector in the previous steps


inputs
Pen P
Your
journey
starts
with
one
step
Your 0.20 0.20 0.19 0.12 0.12 0.14
journey 0.13 0.23 0.23 0.12 0.10 0.15
starts 0.13 0.23 0.23 0.12 0.11 0.15
with 0.14 0.20 0.20 0.14 0.12 0.17
one 0.15 0.19 0.19 0.13 0.18 0.12
step 0.13 0.21 0.21 0.14 0.09 0.18
This row contains the attention weights (normalized attention scores) computed previously